

MUSCLE SHOALS 2 N, AL, USAWMO: **750050**Lat: **34.7728N**Lon: **87.6399W**Elev: **162**StdP: **99.40**Time Zone: **-6.00 (NAC)**Period: **07-19**WBAN: **63894****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -7.9	(c) -5.5	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 7.6	(k) 8.4	(l) 7.2	(m) 11.1	(n) 2.7	(o) 330	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.5	(c) 34.7	(d) N/A	(e) 33.5	(f) N/A	(g) 32.5	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 2.7	(p) 280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 6.7	(b) 6.0	(c) 5.3	(e) -11.5	(f) 36.3	(g) 2.8	(h) 1.9	(i) -13.5	(j) 37.7	(k) -15.2	(l) 38.8	(m) -16.8	(n) 39.8	(o) -18.8	(p) 41.2		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.3	4.5	7.4	11.4	16.1	20.7	25.1	26.1	25.9	23.1	16.7	10.2	7.4		
	DBStd	8.73	6.08	5.89	5.26	4.14	3.55	2.19	2.03	2.15	3.16	4.54	4.78	5.47		
	HDD10.0	531	191	112	48	4	0	0	0	0	0	3	53	119		
	HDD18.3	1746	429	309	220	91	21	0	0	0	4	86	246	340		
	CDD10.0	2816	21	38	91	187	332	452	499	493	393	212	61	39		
	CDD18.3	989	0	2	5	23	94	202	241	234	147	36	3	1		
	CDH23.3	9206	0	5	40	213	829	1981	2292	2183	1293	353	16	1		
(13)	CDH26.7	3792	0	0	3	27	254	878	1026	992	518	92	2	0		
(14)	Wind	WSAvg	2.3	2.8	2.9	3.0	2.8	2.1	2.0	1.8	1.7	2.1	2.2	2.6		
(15) Precipitation	PrecAvg	1378	115	115	149	112	128	115	113	85	104	78	116	143		
	PrecMax	2001	299	273	362	325	301	341	366	182	257	210	281	487		
	PrecMin	995	15	13	34	9	21	26	23	4	2	0	28	16		
	PrecStd	278	59	59	79	65	65	66	69	49	60	56	52	91		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.0	24.1	27.0	29.0	32.3	35.7	35.9	37.9	34.9	31.4	25.2	22.9		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	19.0	21.6	24.3	27.5	30.7	33.8	34.1	34.9	33.2	29.1	22.5	20.6		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	16.8	19.7	22.2	26.0	29.4	32.6	32.9	33.2	31.7	27.2	20.8	18.7		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(26) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	DB	14.6	17.2	20.3	24.2	28.1	31.4	31.8	31.7	29.9	25.2	18.9	16.5		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(34) Mean Daily Temperature Range	5% DB	MDBR	10.4	10.6	11.4	12.3	11.5	11.7	10.5	11.0	11.8	12.7	12.5	9.7		
		MCDBR	12.3	12.8	13.6	13.7	13.4	13.4	12.2	13.3	13.7	15.0	14.0	10.7		
	5% WB	MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.318	0.326	0.365	0.398	0.438	0.465	0.507	0.521	0.428	0.373	0.338	0.317		
	taud		2.496	2.470	2.386	2.290	2.225	2.174	2.083	2.032	2.284	2.417	2.459	2.516		
	Ebn at Noon		889	921	906	888	853	826	788	769	831	854	855	868		
	Edh at Noon		84	96	113	131	142	149	162	168	125	100	87	78		
(44) All-Sky Solar Radiation	RadAvg		2.44	2.96	4.11	5.29	5.87	6.28	6.00	5.54	4.81	3.83	2.79	2.08		
	RadStd		0.19	0.37	0.37	0.35	0.44	0.48	0.33	0.31	0.41	0.50	0.29	0.21		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (28 neighbors)		+0.45	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+140	+101

Nomenclature: See separate page